Hackathon Challenge: Inventory Optimization for TBO.com

Problem Statement

How can data-driven intelligence help TBO.com maximize availability by making smarter travel inventory procurement decisions?

Build an intelligent system that analyzes historical bookings, demand trends, sold outs , look to book , seasonality and external signals to identify inventory to target to maximize expected profit.

The solution should optimize budget allocation across hotels, flights, destinations, and travel periods , while accounting for demand fluctuations, pricing changes, cancellations, and unsold inventory risks.

Basis on this TBO is expected to plan pre buy , supplier integration and pricing to maximise bookable inventory .

Key Objectives

- **Predict Demand:** Forecast travel demand using booking trends, search data, and external market signals.
- **Identify Opportunities:** Discover high-potential inventory, destination , travel time and favorable procurement prices.
- **Optimize Investments:** Allocate a given budget to maximize expected returns.
- **Maximize Profitability:** Recommend optimal purchase timing, inventory quantities, and selling prices.
- **Manage Risk:** Minimize potential losses from uncertain demand and unsold inventory.

Expected Outcome

Develop a decision-support platform that transforms travel and market data into actionable inventory investment recommendations, helping TBO.com **buy the right inventory, at the right time, within the right budget, to maximize risk-adjusted profits, System will be provided with maximum input budget.**